Bayesian Techniques in Macroeconomics

Modern Business Cycle Theory

Session 1 – Measurement, expectations and the equilibrium real rate

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Master – Quantitative Economics – Paris-Dauphine – PSL

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Dauphine | PSL

Today's session

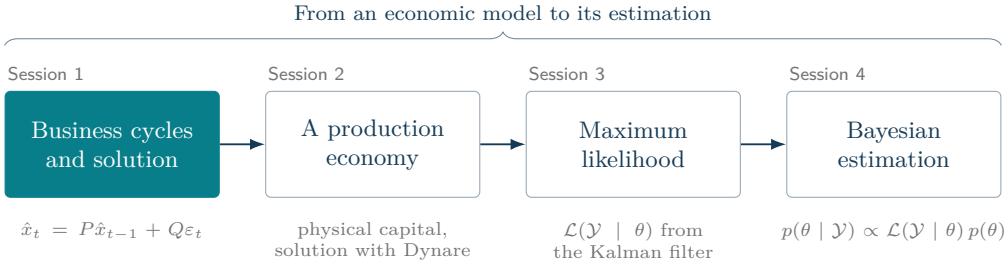
What you already know...

- GDP grows over time and sometimes falls in recessions.
- A consumer equates a marginal rate of substitution to a relative price.
- A Lagrangian turns a constrained choice into first-order conditions.

What we will see:

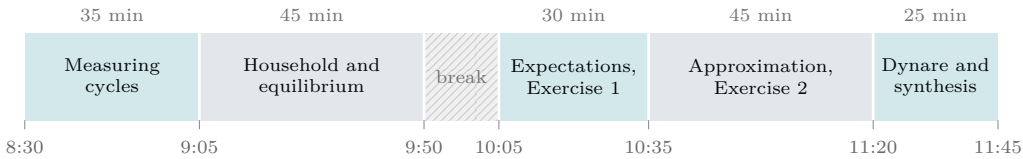
- Measure a business cycle and see why the filter matters.
- Derive the Euler equation of a household that can save.
- Let market clearing select the equilibrium real rate.
- Linearize the model and write its decision rule $\hat{x}_t = P\hat{x}_{t-1} + Q\varepsilon_t$.

Four sessions, one pipeline



- Today: the economic objects that every later step reuses, an equilibrium and its decision rule.
- Bayesian estimation will need a model written as a linear state-space system.

180 minutes of work, one break



- Two exercise blocks, 25 minutes in total, are individual work on Handout 1, followed by an oral correction.
- MATLAB and Dynare are used from the second block onwards. The scripts are in `session1/codes`.

Plan

1. **What is a business cycle?**
2. Consumption, saving and general equilibrium
3. Approximation methods

A cycle is a common movement, not a clock

Definition: business cycle

"A cycle consists of expansions occurring at about the same time in many economic activities, followed by similarly general recessions, contractions, and revivals which merge into the expansion phase of the next cycle."

Burns and Mitchell (1946)

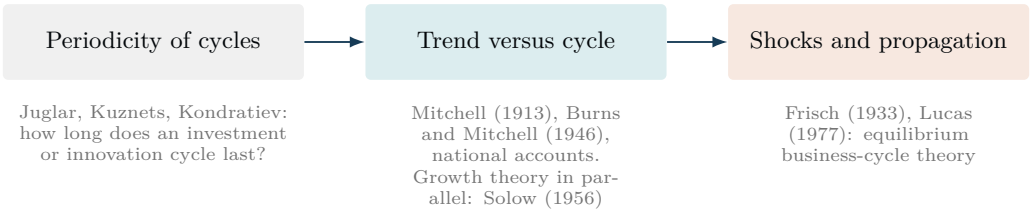
Duration then

From more than one year to ten or twelve years.
Mitchell (1913), Burns and Mitchell (1946)

Duration today

Fluctuations with periods between 6 and 32 quarters. Slower movements are assigned to the trend.

From periodic cycles to trend and cycle



- Before the 1940s, research looked for the period of cycles driven by investment or innovation.
- After Burns and Mitchell (1946), time series are split into a trend and a cycle. Growth theory and fluctuation theory separate.
- This course starts from the business-cycle theory of the 1970s and 1980s.

Why the cycle matters for policy

Dating recessions

NBER and CEPR date the start and end of recessions in the US and in Europe.

After a shock (financial turmoil, pandemic, war), dating and gauging a recession shape the size of fiscal or monetary support.

Measuring the cycle

EU fiscal rules penalize excessive *structural* deficits.

The structural deficit is the observed deficit purged of its cyclical part. A different cycle measure can change the verdict on a country.

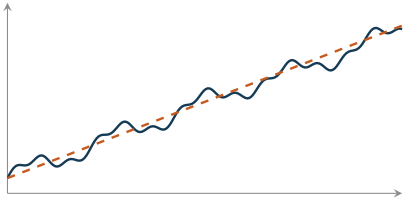
Dating a recession and filtering GDP are related but different exercises.

Trend plus cycle: an identification problem

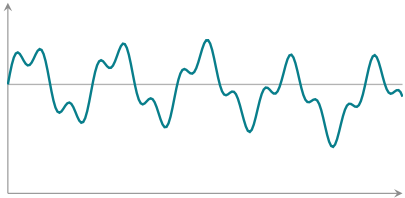
$$y_t = \tau_t + \psi_t$$

τ_t : secular component, ψ_t : cyclical component.

observed y_t and one possible trend τ_t



the implied cycle $\psi_t = y_t - \tau_t$



There is no unique method to separate τ_t from ψ_t. Here a level shift after date 60 becomes a permanently negative "cycle".[1]

[1] Illustrative series, not data.

Logs turn exponential growth into a line

Suppose GDP grows at a constant quarterly rate γ, with disturbances ε_t:

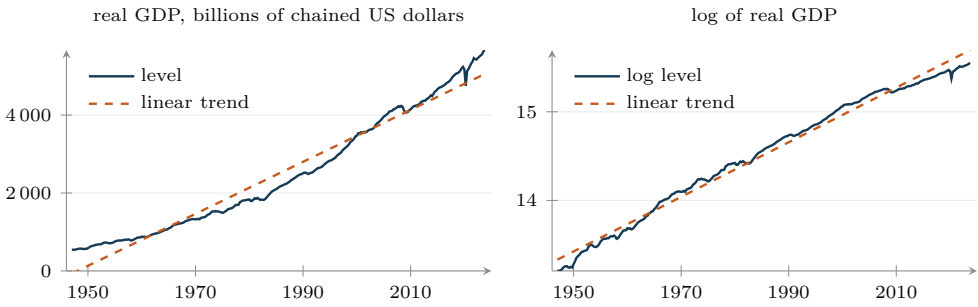
$$Y_t = Y_0 \, (1+\gamma)^t \exp(\varepsilon_t).$$

Taking logs flattens the exponential shape:

$$\log Y_t = \log Y_0 + t \log(1+\gamma) + \varepsilon_t \;\simeq\; \underbrace{y_0 + \gamma \cdot t}_{\text{trend } \tau_t} + \underbrace{\varepsilon_t}_{\text{cycle } \psi_t} .$$

- OLS on $\log Y_t$ estimates the intercept y_0 and the log-growth slope $g = \log(1+\gamma)$.
- The level growth rate is $\gamma = \exp(g) - 1$. For small growth, $g \simeq \gamma$.
- A log deviation of 0.01 is approximately a deviation of one percent.

US real GDP in levels and in logs



- In levels, a straight line misses the exponential shape of GDP.
- In logs, a line is a fair first description, yet slowdowns leave long deviations.[2]

[2] US real GDP, OECD quarterly national accounts via DBnomics (`OECD/QNA/USA.B1_GS1.LNBQRSA.Q`), archived vintage. Script `session1/codes/logGDP.m`.

Three ways to extract the cycle

Linear trend

OLS on $\log Y_t$ with a constant and a time trend.

Simple, conditional on a constant growth rate.

After a structural break, the measured cycle stays negative, as in the US after 2007.

HP filter

A smooth trend that can bend over time, with smoothness $\lambda = 1600$ for quarterly data.

Two-sided: the trend at t uses past and future data. Endpoints move with revisions.

Hodrick and Prescott (1997), Hamilton (2018)

Band-pass filter

Keep the movements with periods between 6 and 32 quarters, remove the rest.

Useful to compare a relationship across frequencies, such as the link between money and inflation.

Baxter and King (1999), Christiano and Fitzgerald (2003)

The HP filter trades fit against smoothness

$$\min_{\{\tau_t\}} \underbrace{\sum_{t=1}^{T}(y_t - \tau_t)^2}_{\text{stay close to the data}} + \lambda \underbrace{\sum_{t=2}^{T-1}\left[(\tau_{t+1} - \tau_t) - (\tau_t - \tau_{t-1})\right]^2}_{\text{penalize changes in trend growth}}$$

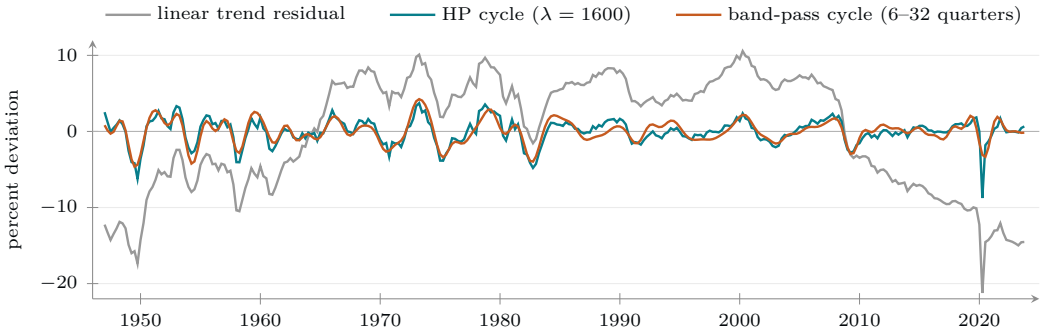
- $\lambda = 0$: the trend is the series itself and the cycle is zero.
- $\lambda \to \infty$: trend growth cannot change, the trend becomes a straight line.
- $\lambda = 1600$ is a convention for quarterly data, not a definition of the business cycle.
- Two-sided filter: adding new quarters revises the last trend estimates.

The band-pass filter keeps a band of periods



- The filter does not fit a trend curve. It retains the frequencies of the chosen band.
- `bpass(gdp,6,32)` returns the cycle. Its complement mixes slow and fast movements: it is not a pure trend.
- With a finite sample the ideal filter must be approximated, poorly near the sample ends.

Three filters, three measured cycles



- HP and band-pass cycles move together. The linear residual shows long swings around a fixed trend.
- Neither $\lambda = 1600$ nor a 6–32 band defines *the* cycle. State the economic question first.[3]

[3] Same series as before, log levels multiplied by 100. Script `session1/codes/BSmeasures.m`.

Reproduce the three filters

```
gdp   = log(mydata(:,2));                % log real GDP
y_lin = gdp-detrend(gdp,1);              % linear trend
y_hp  = hpfilter(gdp,'Smoothing',1600);  % HP trend
c_bp  = bpass(gdp,6,32);                 % 6-32 quarter cycle
```

1. Run `logGDP.m`, then `BSmeasures.m`, from `session1/codes`.
2. Mark which outputs are trends and which are cycles: the HP call returns the *trend*, `bpass` returns the *cycle*.
3. Compare standard deviations and the final quarters of the sample, in percent.

Report series, vintage, sample and filter with every cycle you show.[4]

[4] Internet access is needed for the DBnomics request. HP filtering uses the Econometrics Toolbox.

Small impulses, persistent fluctuations



"Fluctuations in economic activity are seen as the result of small, white noise shocks that affect the economy through a complex dynamic propagation system." Blanchard and Watson (1986)

- The impulse–propagation view goes back to Frisch (1933). Large swings can follow small shocks.[5]
- The old Keynesian view read fluctuations as imbalances in money or labor markets.

[5] **Small shocks can also trigger deep crises:** Kiyotaki and Moore (1997), Boissay, Collard and Smets (2016).

"All business cycles are alike"

	Regularities listed by Lucas (1977)
1	Output co-moves across sectors.
2	Durables are more volatile than non-durables.
3	Prices are procyclical.
4	Short-term interest rates are procyclical.
5	Monetary aggregates and velocity are procyclical.

The research programme

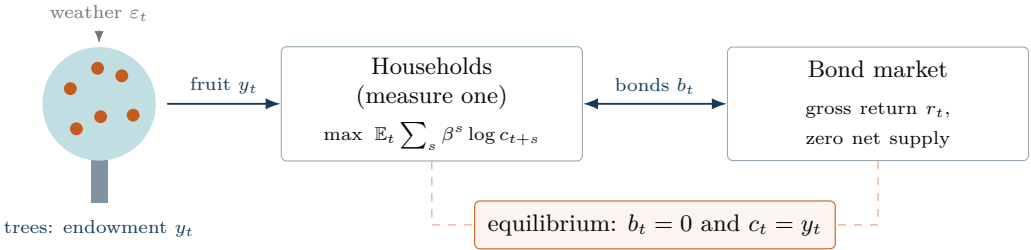
A common mechanism behind fluctuations:

- general equilibrium,
- infinitely lived representative agents,
- rational expectations.

These are historical regularities, not restrictions that every dataset satisfies. Real business cycle theory is one development of this programme.

1. What is a business cycle?
2. **Consumption, saving and general equilibrium**
3. Approximation methods

A Lucas tree economy

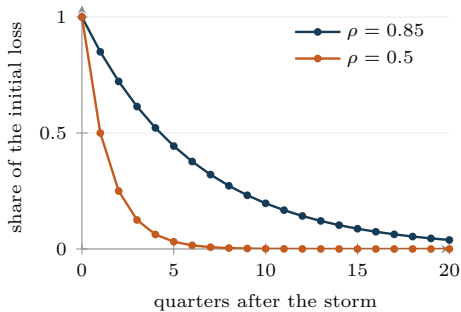


- Fruit falls from trees: output is exogenous. Households can only decide when to eat it.
- A risk-free bond lets one household shift consumption across dates. Fruit is the numeraire.
- The question: which real rate makes zero net saving optimal? Lucas (1978)

Weather shocks with persistence

$$y_t = 1 - \rho + \rho\, y_{t-1} + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, \sigma^2)$$

- y_t: fruit endowment, with steady state $\bar{y} = 1$.
- $0 \leq \rho < 1$: persistence. σ: standard deviation of the innovation.
- A storm damages a branch. It takes a few periods to heal, so a bad draw today also lowers expected fruit tomorrow.



Additive Gaussian shocks are a local teaching device: they allow non-positive fruit with positive probability.

Preferences: patience and concavity

$$\max_{\{c_{t+s},\, b_{t+s}\}_{s\geq 0}} \mathbb{E}_t \sum_{s=0}^{\infty} \beta^s \log c_{t+s}, \qquad 0 < \beta < 1.$$

Discounting β

A lower β puts less weight on future utility: the household is more impatient.

Concave utility

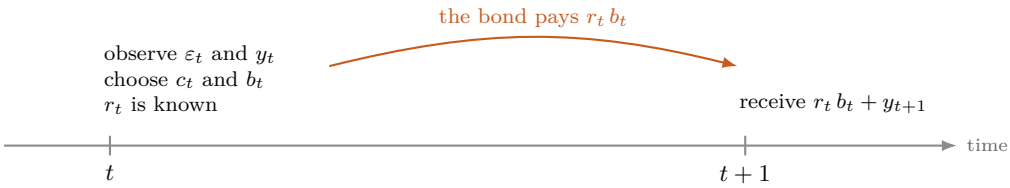
With $\log c$, one more unit of fruit is worth more when consumption is low.

Expectation $\mathbb{E}_t$

Conditional on information at t, including today's fruit. Future shocks remain unknown.

The household chooses consumption c_t and bond holdings b_t in every period and every state.

The budget constraint and its timing



- r_t is the gross real return, known at t and paid at $t+1$. The net real rate is $r_t - 1$.
- The constraint binds because utility increases with consumption.
- A no-Ponzi condition rules out borrowing forever to finance consumption.

The household's Lagrangian

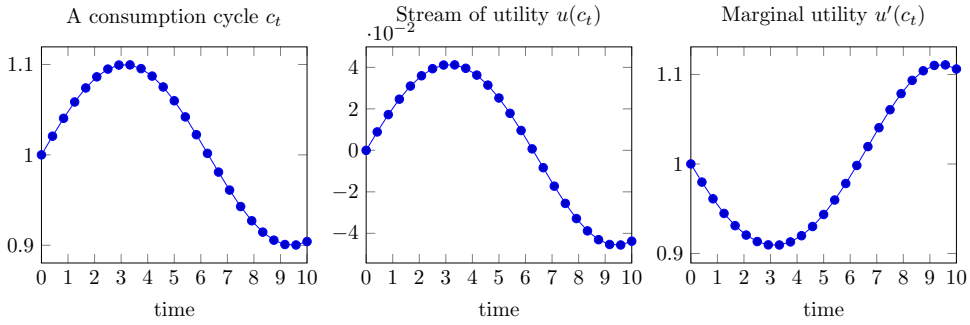
$$\mathcal{L}_t = \mathbb{E}_t \sum_{s=0}^{\infty} \beta^s \Big[\log c_{t+s} - \lambda_{t+s} \big(c_{t+s} + b_{t+s} - r_{t+s-1} b_{t+s-1} - y_{t+s} \big) \Big]$$

- λ_t: value of one more unit of resources at t, the shadow price of the budget.
- Differentiate with respect to c_t and b_t. The bond b_t appears at t and at $t+1$.

$$(c_t): \quad \lambda_t = \frac{1}{c_t} \qquad\qquad (b_t): \quad \lambda_t = \beta\, r_t\, \mathbb{E}_t\{\lambda_{t+1}\}$$

Together with feasibility and transversality, these conditions characterize an interior optimum.

Marginal utility moves against consumption



- Utility rises with consumption, but each extra unit adds less than the previous one.
- A thirsty cyclist values the first glass of water most. When consumption peaks, $\lambda_t = 1/c_t$ is at its trough.[6]

[6] Illustrative consumption cycle under log utility, Handout 1, Figure 1.

The Euler equation

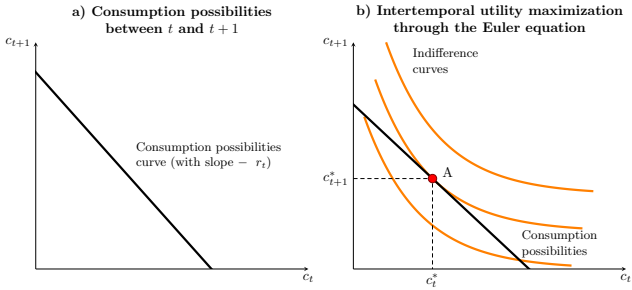
$$\underbrace{u'(c_t)}_{\text{cost of saving one unit today}} = \beta\, r_t \underbrace{\mathbb{E}_t\{u'(c_{t+1})\}}_{\text{expected value of its return tomorrow}}$$

Definition: stochastic discount factor

$m_{t+1} = \beta\, u'(c_{t+1})/u'(c_t)$, the value of one unit of fruit tomorrow in units of fruit today. The Euler equation reads $\mathbb{E}_t\{m_{t+1}\} = 1/r_t$.

- At the optimum, shifting one unit across dates leaves the household equally well off.
- Holding lifetime resources fixed, a higher r_t is an incentive to postpone consumption.

Two dates: a tangency



- Saving one unit today buys r_t units tomorrow: the budget line has slope $-r_t$.
- Indifference curves connect pairs (c_t, c_{t+1}) with equal utility.
- At the consumption point A,

$$r_t = \frac{u'(c_t)}{\beta\, u'(c_{t+1})}.$$

Under uncertainty, the relevant forecast is tomorrow's *marginal utility*, not tomorrow's consumption.[7]

[7] Deterministic two-date diagram, Handout 1, Figure 2.

Market clearing selects the real rate

- No outside lender and identical households: bonds are in zero net supply.
- In equilibrium, each household holds no bond: $b_t = 0$.
- The budget then gives $c_t = y_t$: everyone eats the fruit.
- Households still choose optimally. The rate adjusts so that they choose not to trade.

The equilibrium in two equations

$$\begin{cases} y_t = 1 - \rho + \rho\, y_{t-1} + \varepsilon_t \\ r_t = \left[\beta\, y_t\, \mathbb{E}_t\{y_{t+1}^{-1}\}\right]^{-1} \end{cases}$$

Fruit fixes the allocation. The Euler equation prices the bond.

Expectations change decisions today

Wage setting

Monetary policy accommodates today
→ workers expect higher inflation
→ they ask for a higher wage today.

This led to a new reading of the Phillips curve.

Open-economy portfolios

The domestic rate rises
→ the currency appreciates now
→ then depreciates as expected.

The expected exchange-rate loss offsets the higher return.

- A purely backward-looking model misses both mechanisms.
- Lucas critique: policy changes can change expectations and the decision rules estimated on past data. Lucas (1976)

Rational expectations

Definition: rational expectations

Forecasts consistent with the model and with the information available to agents.

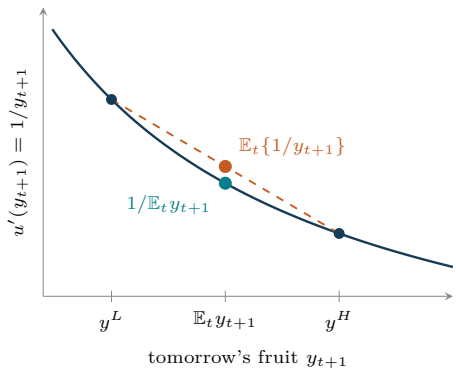
Muth (1961)

$$\begin{aligned}\mathbb{E}_t\{y_{t+1}\} &= \mathbb{E}_t\{1 - \rho + \rho\, y_t + \varepsilon_{t+1}\} \\ &= 1 - \rho + \rho\, y_t,\end{aligned}$$

$$y_{t+1} - \mathbb{E}_t\{y_{t+1}\} = \varepsilon_{t+1}.$$

- The household knows the law of motion and today's fruit, not tomorrow's weather.
- Forecast errors have conditional mean zero.
- Expectations are correct on average, wrong in almost every period.

Risk raises expected marginal utility



$$\mathbb{E}_t\left\{\frac{1}{y_{t+1}}\right\} \geq \frac{1}{\mathbb{E}_t\{y_{t+1}\}}$$

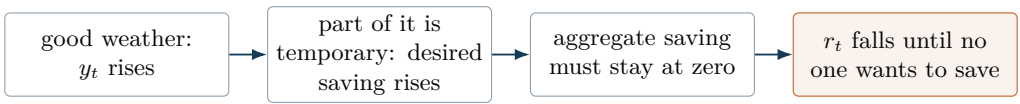
- $1/y$ is convex: risk raises expected marginal utility.
- Certainty equivalence replaces y_{t+1} by its forecast:

$$r_t^{\text{CE}} = \frac{1-\rho+\rho\, y_t}{\beta\, y_t}.$$

- Correct to first order around the steady state, silent on risk.

More fruit today lowers the real rate

$$r_t^{\mathrm{CE}} = \frac{\rho}{\beta} + \frac{1-\rho}{\beta\, y_t}, \qquad \frac{\partial r_t^{\mathrm{CE}}}{\partial y_t} = -\frac{1-\rho}{\beta\, y_t^2} < 0.$$



- With ρ close to one, tomorrow's fruit rises almost as much as today's: the rate barely moves.
- With $\rho = 0$, the whole gain is temporary: the rate falls the most.

Exercise 1: the certainty-equivalent model

Handout 1, Exercise 1 — 15 minutes

Q1–2 Simulate the certainty-equivalent system over 400 periods with $\beta = 0.99$, $\sigma = 0.1$, $\rho = 0.85$: steady state, `randn()` draws, `nan()` vectors, recursive loop. Plot.

Q3 Standard deviation and first-order autocorrelation of output. Compare $\log y_t$ with the log-GDP residual from the linear trend, in the same units.

Q4 Adjust σ and ρ to match US data.

Q5–6 Simulate an impulse response. Compare a large expansion and a large depression.

Q7 Histograms of output and of the real rate. Why do their asymmetries differ?

Check the steady state, the timing of the innovation and the sign of y_t before interpreting any rate.

Correction: persistence versus volatility

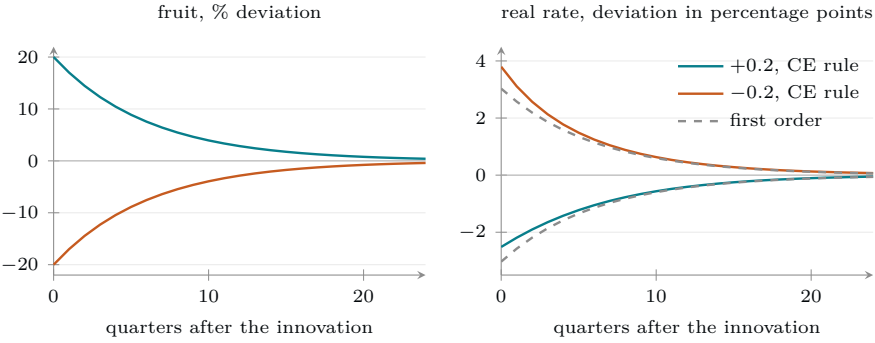

```
for t = 2:T+1
    y(t) = 1-rho+rho*y(t-1)+e(t-1); % fruit
end
r = (1-rho+rho*y)./(beta*y);        % CE rate
```

$$\mathrm{Var}(y_t) = \frac{\sigma^2}{1-\rho^2}$$

$$\mathrm{Corr}(y_t, y_{t-1}) = \rho$$

- σ scales the volatility, ρ governs persistence and amplifies volatility.
- With $\sigma = 0.1$ and $\rho = 0.85$, the standard deviation of y_t is about 0.19, that is 19% of steady-state fruit.
- Match units and filters before comparing with the GDP cycle. These parameters are illustrative, not estimates.

Booms and slumps are not symmetric



- A slump raises the rate more than an equal boom lowers it: r^{CE} is convex in y_t.
- The first-order rule (dashed) is symmetric by construction.[8]

[8] Computed from the displayed equations with $\beta = 0.99$, $\rho = 0.85$ and an innovation of ± 0.2 at date 0. Not empirical responses.

Risk enters through marginal utility

Definition: CRRA utility

$u'(c) = c^{-\sigma_C}$ with curvature $\sigma_C > 0$. Do not confuse σ_C with the shock standard deviation σ. Log utility is $\sigma_C = 1$.

Exact pricing

$$r_t = \frac{y_t^{-\sigma_C}}{\beta\, \mathbb{E}_t\{y_{t+1}^{-\sigma_C}\}}$$

Integrates future marginal utility over its whole distribution.

Certainty equivalence

$$r_t^{\mathrm{CE}} = \frac{1}{\beta}\left(\frac{\mathbb{E}_t\{y_{t+1}\}}{y_t}\right)^{\sigma_C}$$

Captures first-order movements, not risk corrections.

With additive Gaussian fruit, $\mathbb{E}_t\{y_{t+1}^{-\sigma_C}\}$ is not well defined. The exact formula needs positive fruit, as in the Dynare variant at the end of the session.

1. What is a business cycle?
2. Consumption, saving and general equilibrium
3. **Approximation methods**

A model is a set of restrictions

$$\mathbb{E}_t\left[f_\theta(x_{t+1}, x_t, x_{t-1}, \epsilon_t)\right] = 0$$

- x_t: the n endogenous variables, here (y_t, r_t).
- ϵ_t: the p exogenous shocks.
- θ: the deep parameters, here (β, ρ, σ).
- f_θ: the equations of the model.

Definition: decision rule

A function g such that

$$x_t = g(x_{t-1}, \epsilon_t)$$

satisfies the model's equations at every date and in every state.

Solving the model means finding g. Simulations and, in Session 3, the likelihood are computed from g.

The unknown is a function

Under rational expectations, $\mathbb{E}_t\{x_{t+1}\} = \mathbb{E}_t\{g(x_t, \epsilon_{t+1})\}$. Substitute the rule once:

$$\mathbb{E}_t\Big[f_\theta\big(g(x_t, \epsilon_{t+1}),\, x_t,\, x_{t-1},\, \epsilon_t\big)\Big] = 0.$$

Substitute again to express everything in terms of x_{t-1}:

$$\mathbb{E}_t\Big[f_\theta\big(g(g(x_{t-1}, \epsilon_t), \epsilon_{t+1}),\, g(x_{t-1}, \epsilon_t),\, x_{t-1},\, \epsilon_t\big)\Big] = 0.$$

- g must satisfy this restriction for infinitely many states x_{t-1} and shocks ϵ_t.
- Except in a few cases, g is approximated. Setting $\epsilon_{t+1} = 0$ inside a nonlinear expression would discard risk.

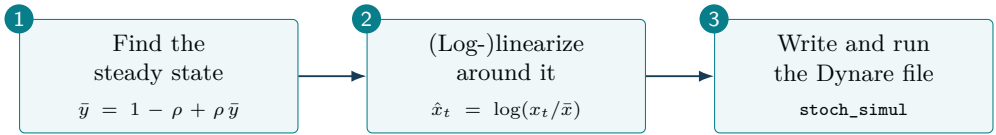
The toy model has an explicit rule

For the certainty-equivalent model, substitute today's fruit into the rate:

$$y_t = 1 - \rho + \rho\, y_{t-1} + \varepsilon_t,$$
$$r_t^{\mathrm{CE}} = \frac{\rho}{\beta} + \frac{\beta^{-1}(1-\rho)}{1 - \rho + \rho\, y_{t-1} + \varepsilon_t}.$$

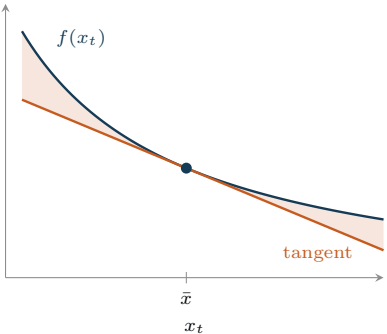
- Both variables are functions of yesterday's fruit y_{t-1} and today's shock ε_t: this is g.
- Yesterday's rate plays no role. It is not a state variable.
- With capital or other state variables, no such closed form exists. We approximate g locally.

Three steps to solve a model



- Many methods solve the linear system: Schur decomposition, undetermined coefficients...
- Under the conditions of Blanchard and Kahn (1980), the linear model has a unique stable solution.

Linearize: replace a curve by its tangent



$$f(x_t) - f(\bar{x}) \simeq f'(\bar{x})\,(x_t - \bar{x})$$

Definition: log deviation

$\hat{x}_t = \log(x_t/\bar{x})$, in decimal units. $100\,\hat{x}_t$ is an approximate percentage deviation from the steady state.

- Since $x_t - \bar{x} \simeq \bar{x}\,\hat{x}_t$:

$$f(x_t) - f(\bar{x}) \simeq f'(\bar{x})\,\bar{x}\,\hat{x}_t.$$

- The shaded gap is the approximation error. It grows with the distance to $\bar{x}$.

Six rules to log-linearize quickly

Level	First-order deviation
x_t	$\bar{x}\,\hat{x}_t$
$x_t\,y_t$	$\bar{x}\,\bar{y}\,(\hat{x}_t + \hat{y}_t)$
x_t / y_t	$(\bar{x}/\bar{y})\,(\hat{x}_t - \hat{y}_t)$
x_t^a	$\bar{x}^a\,a\,\hat{x}_t$
$\log x_t$	$\hat{x}_t$
$\exp x_t$	$\exp(\bar{x})\,\bar{x}\,\hat{x}_t$

With two variables, keep both marginal effects:

$$f(x_t, y_t) - f(\bar{x}, \bar{y}) \simeq f'_x\,\bar{x}\,\hat{x}_t + f'_y\,\bar{y}\,\hat{y}_t.$$

- Each rule gives the expression minus its steady-state value.
- A sum keeps the steady-state weights of its terms.
- A zero-mean shock stays in levels: its log is not defined.

Step 1: the deterministic steady state

Remove time subscripts and set the shock to zero:

$$\begin{cases} \bar{y} = 1 - \rho + \rho\,\bar{y} \\ \bar{r} = \dfrac{1 - \rho + \rho\,\bar{y}}{\beta\,\bar{y}} \end{cases} \qquad \Longrightarrow \qquad \boxed{\bar{y} = 1, \qquad \bar{r} = \frac{1}{\beta}}$$

- With $\beta = 0.99$, the gross quarterly rate is $\bar{r} \simeq 1.0101$, a net rate of about 1% per quarter.
- Impatience alone sets the steady-state rate.
- The deterministic steady state is the expansion point, not the stochastic mean of a nonlinear variable.

Step 2: linearize each equation

Endowment

$$y_t - \bar{y} = \rho\,(y_{t-1} - \bar{y}) + \varepsilon_t$$
$$\bar{y}\,\hat{y}_t \simeq \rho\,\bar{y}\,\hat{y}_{t-1} + \varepsilon_t$$
$$\hat{y}_t \simeq \rho\,\hat{y}_{t-1} + \varepsilon_t$$

The constant disappears. The shock stays in levels because its steady state is zero.

Real rate

$$r_t = \frac{1-\rho}{\beta}\,\frac{1}{y_t} + \frac{\rho}{\beta}$$
$$\bar{r}\,\hat{r}_t \simeq -\frac{1-\rho}{\beta\,\bar{y}}\,\hat{y}_t$$
$$\hat{r}_t \simeq -(1-\rho)\,\hat{y}_t$$

Use $\bar{y} = 1$ and $\bar{r} = 1/\beta$ in the last step.

A 1% rise in fruit lowers the gross rate by about $(1-\rho)\%$ of its steady state. In net percentage points, use $100\,\bar{r}\,\hat{r}_t$.

The linear Euler equation

$$\underbrace{-\hat{y}_t}_{\text{marginal utility today}} = \hat{r}_t \underbrace{- \mathbb{E}_t \hat{y}_{t+1}}_{\text{expected marginal utility}} \implies \boxed{\hat{r}_t = \mathbb{E}_t \hat{y}_{t+1} - \hat{y}_t}$$

- With log utility, marginal utility moves one-for-one against consumption, here fruit.
- Rational expectations give $\mathbb{E}_t \hat{y}_{t+1} = \rho\, \hat{y}_t$, hence $\hat{r}_t = -(1-\rho)\, \hat{y}_t$.
- The rate is high when fruit is expected to grow: households would like to borrow against it.
- Risk corrections disappear at first order, not from the economic problem.

The decision rule in state-space form

Substitute the endowment into the rate to make the system recursive:

$$\underbrace{\begin{bmatrix}\hat{y}_t \\ \hat{r}_t\end{bmatrix}}_{\hat{x}_t} = \underbrace{\begin{bmatrix}\rho & 0 \\ -(1-\rho)\,\rho & 0\end{bmatrix}}_{P} \underbrace{\begin{bmatrix}\hat{y}_{t-1} \\ \hat{r}_{t-1}\end{bmatrix}}_{\hat{x}_{t-1}} + \underbrace{\begin{bmatrix}1 \\ -(1-\rho)\end{bmatrix}}_{Q} \varepsilon_t$$

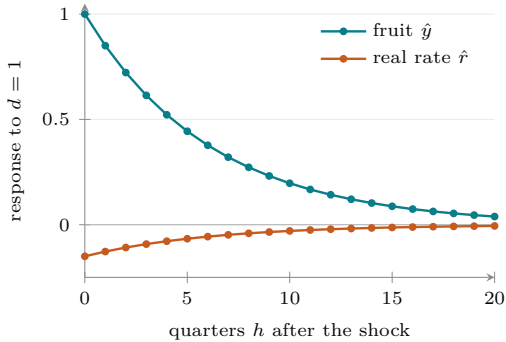
- The second column of P is zero: the lagged rate is not a state.
- The eigenvalues of P are ρ and 0. With $|\rho| < 1$, every response dies out.
- Knowing P and Q, the shocks and the initial state, we can generate the whole path $\{\hat{y}_t, \hat{r}_t\}$.

An impulse response isolates propagation

$$\hat{x}_0 = 0, \quad \varepsilon_1 = d, \quad \varepsilon_t = 0 \ (t > 1)$$

$$\Longrightarrow \quad \boxed{\hat{x}_{1+h} = P^h Q\, d}$$

- The impact response is $Q\,d$. Each quarter multiplies by P.
- Doubling d doubles every response.
- Changing the sign of d flips every response: linear IRFs are symmetric.



Exercise 2: the linear model

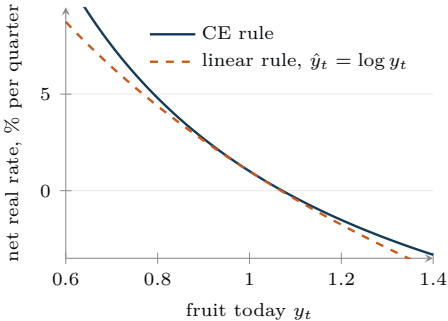
Handout 1, Exercise 2 — 10 minutes

- Q1–2 Simulate the linear system over 400 periods, in the same script as Exercise 1. Plot.
- Q3 Simulate an IRF: one shock at the first period, then no shock.
- Q4 Does a large expansion move the real rate as much as a large depression?
- Q5 Histograms of the linearized variables. Why are they symmetric here?
- Q6 Mean squared error $\mathbb{E}\big[(r_t - \bar{r}(1+\hat{r}_t))^2\big]$ for the gross rate, with the same innovations as in Exercise 1.

Compare the two rates in levels. Never compare a gross-rate level with its log deviation.

Correction: the linear rule and its error


```
P = [rho 0; -(1-rho)*rho 0];  % transition
Q = [1; -(1-rho)];            % impact
x = zeros(2,T+1);             % log deviations
for t = 2:T+1
    x(:,t) = P*x(:,t-1)+Q*e(t-1);
end
r_lin = (1/beta)*(1+x(2,:)'); % gross rate
mse   = mean((r-r_lin).^2);   % error
```



$\partial^2 r_t^{\mathrm{CE}}/\partial y_t^2 = 2(1-\rho)/(\beta\, y_t^3) > 0$: the tangent misses the curvature. Smaller shocks, smaller error.

Dynare: a positive-endowment variant


```
model;
y = y(-1)^rho*exp(e);
1 = beta*r*(y/y(+1))^sigma;
end;
```

- Fruit is now lognormal: always positive, with finite inverse moments.
- `sigma` is the CRRA curvature σ_C, `sigy` the shock standard deviation.

An exact risk correction

With $y_{t+1} = y_t^{\rho} \exp(\varepsilon_{t+1})$,

$$\mathbb{E}_t\{y_{t+1}^{-\sigma_C}\} = y_t^{-\rho\sigma_C} \exp\!\left(\sigma_C^2\sigma^2/2\right),$$

$$r_t = \beta^{-1}\, y_t^{\sigma_C(\rho-1)} \exp\!\left(-\sigma_C^2\sigma^2/2\right).$$

More risk lowers the risk-free rate: households save as a precaution. The effect is second order in the shock size.

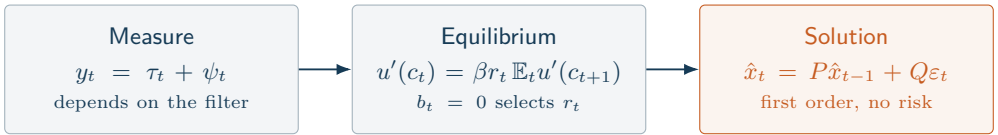
Run and check the Dynare file


```
steady;
resid;
check;
stoch_simul(order=1,irf=30) y r;
```

1. Add Dynare to the MATLAB path and move to `session1/codes`.
2. In a working copy of `dyn_toy_model.mod`, set `sigma=1` and `order=1`.
3. Run `dynare dyn_toy_model` and compare the IRFs with your linear rule.
4. Then return to `order=2` to see the risk correction, without changing the endowment process.

File	Endowment	Rate
`toy_model.m`	additive Gaussian, 10,000 periods	CE rule, log utility
`toy_model_bootstrap.m`	additive Gaussian, Monte Carlo draws	simulated expectation
`dyn_toy_model.mod`	lognormal, $\sigma_C = 10$	order-2 perturbation

What to retain



- A measured cycle depends on trend and frequency conventions.
- Rational expectations use the conditional distribution, not perfect foresight.
- Certainty equivalence and first-order approximation keep the mechanisms, not every risk effect.
- The decision rule is the bridge to simulation and, in Session 3, to the likelihood.

Next session: output becomes endogenous through investment in physical capital.

After class

- Read Handout 1, §1.1–1.3, for the history and the caveats of filtering.
- Rebuild the Euler equation and the equilibrium of §2.1–2.5 without the slides.
- Finish Exercise 1 (questions 1–7) and Exercise 2 (questions 1–6). Bring your comparison of the two rates under common shocks.
- Use §3.1–3.3 to separate economic equilibrium, local approximation and recursive solution.

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